Clinics in Dermatology (2017) **35**, 285–291

Clinics in Dermatology

Use of hypnosis, meditation, and biofeedback in dermatology

Philip D. Shenefelt, MD*

Department of Dermatology and Cutaneous Surgery, University of South Florida College of Medicine, Tampa, Florida

Abstract Hypnosis utilizes trance to access otherwise inaccessible repressed or unconscious memories and features of the psyche and control of physiology not attainable in the ordinary conscious waking state. Medical uses of hypnosis in dermatology include reducing discomfort from itching or skin pain, altering ingrained dysfunctional habits such as scratching, promoting healing of skin disorders, searching for psychosomatic aspects of skin disorders and alleviating them, and reframing cognitive and emotional dysfunctional patterns related to skin disorders. Meditation uses trance to center and balance. Medical uses of meditation in dermatology include relaxation to promote healing of skin disorders and refocusing with respect to the meaning and emotional negative valance of skin disorders. Biofeedback in dermatology employs instrumentation with visual or auditory feedback to permit conscious awareness and alteration of physiologic phenomena such as sweating as measured by galvanic skin resistance and skin temperature measured by temperature detecting devices, promoting relaxation and healing. These methods and techniques permit access to and intervention in otherwise inaccessible areas that can influence skin disorders. With proper use, they are very safe, with minimal, if any, side effects and sometimes produce significant results where other methods have failed.
© 2017 Elsevier Inc. All rights reserved.

Hypnosis, meditation, and biofeedback provide tools for access to aspects of psyche and physiology not ordinarily accessible in the ordinary conscious waking state. Hypnosis and meditation use natural trance states, whereas biofeedback uses instrumentation with visual or auditory feedback of physiologic phenomena. These provide additional avenues for assessment of and intervention in skin conditions and disorders.

The brain cycles through diurnal (daily) and ultradian (roughly every 90 minutes) phases of alertness and less alertness.[1] Higher gamma (38-90 Hertz [Hz] or cycles per second) frequencies help to integrate thoughts and complex motor processes across the brain. Predominant brain wave frequencies corresponding to hypervigilance, fright, and anxiety are in the high beta (18-38 Hz) range; those corresponding to focused alertness are in the beta (12-18 Hz) range; those corresponding to relaxed calm alertness, mindfulness meditation, and alert awake hypnosis are in the alpha (8-12 Hz) range; those corresponding to rapid eye movement sleep, internally focused meditation, and hypnotic trance are in the theta (4-8 Hz) range; and those corresponding to deep sleep are in the delta (0.5-4 Hz) range. The ultradian cycling while awake is between beta focused alert and alpha relaxed and calm and while asleep is between theta dreaming rapid eye movement sleep and delta deep sleep.

* Corresponding author. Tel.: +1 813 974 4270; fax: +1 813 974 4272.
E-mail address: pshenefe@health.usf.edu.

http://dx.doi.org/10.1016/j.clindermatol.2017.01.007
0738-081X/© 2017 Elsevier Inc. All rights reserved.

Trance

Trance is a nonsleep shift into a low alpha or theta range that occurs naturally, such as when deeply absorbed in a story, book, movie, or thought (parasympathetic dominance). Trance can also occur during repetitive strong activity, such as jogging or intense dancing (whirling dervish), and during overwhelming emotional shock or physical trauma (sympathetic overdrive). Trance can also be induced intentionally. The hypnotic trance has objectively documented differences in regional cerebral blood flow[2] and electroencephalography patterns[3] compared with the usual waking state. Our shamanic and religious ancestors induced trance by drumming, rattling, dancing, fasting, chanting, praying, or storytelling. For meditation, trance is usually induced through concentration on slow breathing, chanting, or a visual mandala, whereas for hypnosis many types of inductions have been described, such as eye fixation, eye roll, progressive relaxation, and many others. Meditation evolved from Eastern religious traditions focusing on centering and balance and often nonattachment, whereas hypnosis evolved from Western religious traditions focusing on exorcism and became secularized using the Western fix-it interventional approach to life. Both meditation and hypnosis currently have secular health care applications, including specific uses in dermatology.

Hypnosis

Precisely defining consciousness has been as elusive as precisely defining hypnosis. Most regulation of body functioning and activity is unconscious and rapid, based on physiologic functioning and habitual neural patterns. Conscious functioning and activity is much more limited in scope and comparatively much slower. By comparison, if the unconscious is compared with the size of a cantaloupe and functions at supercomputer speed, the conscious is the size of a walnut and functions at slow, first generation personal computer speed. Habit thus usually wins out in the long run over willpower. Changing a dysfunctional habit by repeatedly doing something different consistently for 20 to 40 times helps to establish a new habit. The new habit then begins to occur automatically in place of the old dysfunctional habit.

Using hypnosis in dermatology

Some patients are already in trance at the office visit, and the task then is to discern and use it. The trance state may also be induced by using deep slow breathing, relaxation, guided imagery, or other hypnosis induction techniques. Although patients vary in their ability to enter the trance state, most can obtain some benefit from hypnosis. Medical hypnotherapy can reduce suffering, promote healing, or help the patient alter a destructive behavior pattern. One author[4] described hypnosis as a psychophysiologic tetrad of altered consciousness consisting of narrowed awareness, restricted and focused attentiveness, selective wakefulness, and heightened suggestibility. For a more detailed discussion of the definitions and theories regarding hypnosis, see a recent textbook on hypnosis.[5]

Many myths and misconceptions about hypnosis tend to overrate, underrate, or distort the true capabilities and nature of hypnosis. For dermatologic issues, hypnotic suggestion may help decrease pain and pruritus in the skin and intervene in psychosomatic aspects of skin diseases. Hypnotic suggestion or psychosomatic hypnoanalysis can lead to the resolution of some skin diseases, including verruca vulgaris (see later). In some cases, suggestion without formal trance induction may be sufficient. One study[6] reported successful use of suggestion to treat verrucae vulgaris.

For skin disorders resistant to direct suggestion, psychosomatic hypnoanalysis may prove fruitful with exploration of seven key factors[7,8]:

1. **C**onflict between want to and ought to
2. **O**rgan language
3. **M**otivation or secondary gain
4. **P**ast traumatic experiences
5. **A**ctive identification with similar issue in a significant person
6. **S**elf-punishment
7. **S**uggestion or imprint

These seven COMPASS factors are common issues that may retard responsiveness to direct suggestion. In occasional instances, there may be a spiritual aspect that impedes healing and can be addressed.[9] Obtaining training in the safe use of hypnosis is no more difficult that learning to do dermatologic surgery safely. The American Society of Clinical Hypnosis (www.asch.net) is a good source for basic, intermediate, and advanced training in medical use of hypnosis. Another option is to refer the patient to a competent medical hypnotherapist, who can be found on the American Society of Clinical Hypnosis website.

Using hypnotic relaxation for dermatologic procedures

Several randomized control trials have reported the efficacy of hypnotic relaxation for procedures. For example, one study[10] found hypnotic relaxation to be effective in a prospective, randomized, controlled trial comparing the efficacy of a local anesthetic (EMLA), EMLA plus hypnosis, or EMLA plus attention in children receiving venipuncture. Children in the EMLA-plus-hypnosis group reported less anticipatory anxiety, less procedure-related pain and anxiety, and less behavioral distress during the procedure than patients in the other two groups. Parents whose children were randomized to the

EMLA-plus-hypnosis group also experienced less anxiety during their child's procedure than parents whose children had been randomized to the other two groups.[10]

In a large prospective randomized trial[11] of adjunctive nonpharmacologic analgesia for invasive percutaneous vascular radiologic procedures, patients were randomized into three groups: standard intravenous conscious sedation care, structured attention with conscious sedation, and guided self-hypnotic relaxation with conscious sedation if needed. The researchers found that pain scores increased linearly with time in the standard group and in the structured attention group but remained flat in the hypnosis group, whereas anxiety scores decreased over time in all three groups but more so in the hypnosis group. Conscious sedation drug use was significantly higher in the standard group, intermediate in the structured attention group, and lowest in the self-hypnosis group. Hemodynamic stability was significantly higher in the hypnosis group than in the other two groups. The lengths of procedure times were significantly shorter in the hypnosis group than in the standard group.[11] Individual imagery was quite varied from one patient to another.[12]

Allowing the patient to choose his or her own self-guided imagery seems to permit most individuals to reach a state of relaxation during procedures, regardless of how hypnotizable they are.[12] The author has used a technique similar to the invasive radiologic studies modified for dermatology with good success in dermatologic surgery.[13] The eye-roll induction was most commonly used by the author, and this method works quickly for most patients. The author[14] conducted a randomized controlled trial of hypnotic relaxation for dermatologic surgery. Randomization was into live hypnotic induction, recorded hypnotic induction, and control groups. The live hypnotic induction group had significantly less anxiety by 20 minutes into the procedure than either the recorded hypnotic induction or the control group. Both the live hypnotic induction and the recorded hypnotic induction groups developed self-guided imagery, whereas the control group did not.

Using hypnosis for specific dermatologic disorders

Acne excoriée

Persistent picking at acne often increases scarring. Posthypnotic suggestion was successfully used in one study, where the patient was to remember the word scar whenever she began reaching up to pick at her acne and to refrain from picking by saying scar. The psychogenic picking of acne excoriée was controlled in two cases.[15] Another case report concluded that this posthypnotic suggestion can be used successfully, along with suggestions that natural beauty with slight imperfections is really more beautiful than something artificially perfect.[16]

Alopecia areata

Stress reduction through hypnotic suggestion may benefit patients with alopecia areata. A strong correlation between high stress reactivity and depression in patients with alopecia areata has also been reported.[17] Another group[18] used hypnotherapy for 21 refractory alopecia areata patients, 9 of whom had alopecia universalis and 12 of whom had extensive alopecia areata. All patients experienced significantly lower anxiety and depression after hypnotherapy. Complete scalp hair regrowth occurred in nine patients, including four with alopecia universalis and two with ophiasis. More than 75% scalp hair regrowth occurred in another three patients. Unfortunately, five patients had a significant relapse of their alopecia.

Atopic dermatitis

Stress reduction and behavior alteration through hypnotic suggestion can benefit patients with atopic dermatitis. A nonrandomized control trial[19] treated 18 adults with extensive and refractory atopic dermatitis with hypnotherapy, and relaxation, stress management, direct suggestion for nonscratching behavior and for skin comfort and coolness, ego strengthening, posthypnotic suggestions, and instruction in self-hypnosis were used. This resulted in statistically significant reduction in itching, scratching, sleep disturbance, and tension compared with controls. The use of topical corticosteroids decreased for each patient by 40% at 4 weeks, 50% at 8 weeks, and 60% at 16 weeks. For resistant atopic dermatitis, hypnosis can also reduce the required amount of other conventional treatments.[20]

Congenital ichthyosiform erythroderma

Several cases of remarkable clearing of congenital ichthyosiform erythroderma of Brocq have occurred following direct suggestion for clearing under hypnosis. One study[21] used suggestion that resulted in complete clearing in one case, where the author had mistakenly thought that he was suggesting away warts and strongly believed in the efficacy of the suggestion. Later he was sent several other cases but did not succeed after knowing the true diagnosis and having less confidence in the expected outcome of the suggestion. Another case study[22] reported improvement in a 34-year-old father and his 4-year-old son.

Dyshidrotic dermatitis

A reduction in the severity of dyshidrotic dermatitis has been reported with hypnotic suggestion.[23] Greisemer's[24,25] index rating of dermatologic disorders on a percentage scale from 0 to 100 based on emotional triggering of the disorder indicates a significant psychosomatic component for dyshidrosis, so hypnotic suggestion has biological plausibility as a therapy.

Erythema nodosum

Resistant erythema nodosum of 9 years' duration resolved in a 44-year-old woman after psychosomatic hypnoanalysis.[26] Five of the seven key COMPASS factors (Conflict, Organ language, Motivation, Past traumatic experience, Active identification, Self-punishment, Suggestion or imprint) were positive in this case. Because so many factors were positive, the patient was referred for psychotherapy.

Erythromelalgia

Successful treatment of erythromelalgia in an 18-year-old woman using hypnosis alone, followed by self-hypnosis, with permanent resolution has been reported.[27]

Furuncles, recurrent

A 33-year-old man, from age 17, had recurrent multiple furuncles that contained *Staphylococcus aureus* and were unresponsive to multiple treatment modalities. It was noted that the patient had a negative self-image and used hypnotic suggestions with imagined sensations of warmth, cold, tingling, and heaviness.[28] This along with self-hypnosis resulted in dramatic improvement over 5 weeks, with full resolution of the recurrent furuncles. The patient also reported substantial improvement of his self-image. The hypnotic suggestions likely helped in some way to normalize the immune response to the bacteria. In unusually resistant cases with significant psychosomatic overlay, hypnotic suggestions may help to end the recurrent cycles of infection.

Glossodynia

Painful tongue may respond to hypnotic suggestions for comfort and pain reduction. The pain source may be either purely organic or have a psychogenic component.[29] If not responsive to direct hypnotic suggestions, psychosomatic hypnoanalysis may be considered.

Herpes simplex infection

The frequency of recurrences of herpes simplex infection has been reduced by hypnotic suggestion along with lessening of discomfort from recurrent herpetic eruptions.[30] In patients with an apparent emotional trigger factor, hypnotic suggestion may be useful as a therapy for reducing the frequency of recurrence and for lessening of discomfort. Resistant cases may be explored with psychosomatic hypnoanalysis.

Hyperhidrosis

Hypnotic suggestion and autogenic training may be useful as adjunctive therapies for treating hyperhidrosis, but the success rate has been variable.[31]

Ichthyosis vulgaris

A 33-year-old man with ichthyosis vulgaris that tended to be better in the summer and worse in the winter began hypnotic suggestion therapy in the summer and was able to maintain the summer improvement throughout the fall, winter, and spring.[32]

Lichen planus

In selected cases of lichen planus, pruritus and lesions may be reduced by using hypnotic suggestion, but results have been mixed.[33]

Neurodermatitis

Cases of neurodermatitis or psychogenic excoriations have resolved by using hypnotic suggestion and remaining resolved with up to 4 years of follow-up.[34] A case series[35] with three patients with neurodermatitis, failing to respond to direct hypnotic suggestion, responded to psychosomatic hypnoanalysis with ideomotor signaling followed by reframing.

Nummular dermatitis

Reduction of pruritus and resolution of lesions of nummular dermatitis after hypnotic suggestions have been reported with mixed results.[33]

Postherpetic neuralgia

The pain of acute herpes zoster and postherpetic neuralgia can be reduced by hypnotherapy, but the results have been mixed.[33] There is a reported case of an older man with debilitating postherpetic neuralgia not responding to 6 years of multiple treatment modalities. He reacted favorably to hypnotic suggestion, along with self-taught hypnosis, and regained a sense of control over his life. He could temporarily lessen his pain and even play nine holes of golf, whereas before he would have to stop after a hole or two. His entire outlook on life changed, and the pain was no longer in control.

Pruritus

The intensity of pruritus may be modified and diminished by hypnotic suggestions.[33] A man with chronic myelogenous leukemia had intractable pruritus that lessened with hypnotic suggestion.[36]

Psoriasis

Excess stress may be considered a significant factor in the onset, exacerbation, and prolongation of psoriasis. Hypnotic suggestions have been reported to have a positive effect on

psoriasis.[37–39] Seventy-five percent clearing of psoriasis was reported by using a hypnotic sensory-imagery technique.[40] In another study, a patient with extensive, severe psoriasis of 20 years' duration experienced marked improvement by using sensory imagery to replicate the sensations in the patient's skin that he had experienced during sunbathing.[41] Another case of severe psoriasis of 20 years' duration fully resolved with a hypnoanalytic technique.[42]

A small randomized, double-blind controlled trial using hypnotic suggestion as adjunctive therapy for psoriasis reported significant improvement in individuals who were highly hypnotizable. The investigators used hypnosis with both the experimental participants and the controls. The experimental participants had suggestions to imagine experiencing a treatment that had worked well for them in the past, whereas the controls had neutral suggestions.

Rosacea

The vascular blush and papular components of rosacea have been reported to improve in selected cases of hypnotic suggestions for resistant rosacea.[43]

Trichotillomania

Successful adjunctive treatment of trichotillomania with hypnosis has been reported.[44] Hypnotic suggestion was used in three adolescents where they become protective patrons of their weak and vulnerable hairs with cessation of hair pulling.[45] Dabney Ewin (personal communication) has had moderately good results in three cases by disrupting the unconscious activity with the suggestion to "let it be impossible to raise your hand above the shoulder towards your hair without first stopping to look at your hand and make a conscious decision about possibly doing something different with your hand."

Urticaria

An 11-year-old boy had an urticarial reaction to chocolate that could be blocked by hypnotic suggestion so that hives appeared on only one side of his face.[46] A case series study of hypnosis with relaxation therapy used for 15 patients with chronic urticaria of an average duration of 7.8 years reported that within 14 months, 6 patients' conditions had cleared and 8 had improved, with decreased medication requirements reported by 80% of patients. One patient's condition did not improve.[47] Psychosomatic hypnoanalysis uncovering a significant history in a medical student who had urticaria after eating chocolate was also described.[7] As a 4-year-old boy he had gone to a zoo with his parents and happened to be at the python cage at feeding time. He was horrified to see a python swallow a live rabbit. The next morning was Easter and he got an Easter basket containing a chocolate bunny. When he ate the chocolate bunny, he developed hives. After the medical student became aware of his own childhood memories and processed them as an adult, along with posthypnotic suggestions that it would be safe for him to eat chocolate without reacting, he could eat chocolate with no reaction.

Verruca vulgaris

Reported efficacy of suggestion in treating warts[6] has since been confirmed numerous times. Case series reports also confirm the efficacy of hypnosis in treating warts.[48,49] Several randomized control trials have reported significant clearing of warts using hypnosis. In a well-conducted randomized controlled study,[50] it was found that 53% of the experimental hypnosis suggestion group had improvement of their warts 3 months after the first of 5 hypnotherapy sessions, whereas none of the control group had improvement. A case series of patients with warts resistant to other measures, including direct hypnotic suggestion, reported that 33 of 41 cleared after psychosomatic hypnoanalysis.[51] In these cases, a psychosomatic factor had been inhibiting resolution of the warts.

Vitiligo

The psychoneuroimmunologic aspects and mechanisms of vitiligo have been described.[52] Occasional cases of vitiligo have improved by using hypnotic suggestion, but most do not[33]; however, the attitude of the person toward accepting the vitiligo may improve.

Using meditation in dermatology

Meditation has long been known as a means of reducing stress. Meditation may be broadly categorized into concentrative meditation where the focus is on one object such as slow breathing, a candle flame or image, mandala, word, or mantra and mindfulness meditation where the focus is on emotional nonattachment but broad awareness of many objects, sounds, other sensations, or thoughts. Both involve entering a trance state. The concentrative trance is associated with reduced external awareness similar to that of an internally focused hypnotic trance, whereas the mindfulness trance maintains external awareness while remaining calmly centered similar to that of an alert awake hypnotic trance.

The Eastern paradigm for healing looks more at the "what" of disease, examining the systems and supersystems involved and the means to restore or rebalance the system with a long-term focus, whereas the Western scientific paradigm for healing generally evaluates the "how" of disease, examining the subsystems involved and the means to repair and cure or control the problem with a short-term focus.[53] Hypnosis arose in the Western cultural milieu in Europe, whereas meditation arose in the Eastern cultural milieu, primarily in India with spread throughout Asia. They both use trance phenomena but with different conceptual approaches and different types of emphasis.

A form of concentrative meditation known as the relaxation response was introduced.[54] It is performed by sitting in a quiet place, closing your eyes, and letting your muscles loosen and relax, starting with your feet and working upward with progressive muscular relaxation trance induction, breathing evenly through your nose and becoming aware of the breath as breath trance induction. With each exhalation, say the word "one" to yourself as a concentrative mantra meditation trance induction. Allow any distracting thoughts or sensations to drift away, ignored like clouds in the sky. Continue the concentrative meditation for 10 to 15 minutes. Afterward, remain sitting quietly for a few minutes, first with your eyes closed, then with your eyes open. The health benefits of the relaxation response have been extensively researched, with positive results in areas such as cardiovascular health.

Mindfulness meditation has also been adapted for medical use for stress reduction. The Mindfulness-Based Stress Reduction Program consisted of an 8-week course with weekly 2-hour classes where techniques of breath, awareness of body sensations, and stretching yoga were taught. The course also included a half day of mindfulness meditation and daily homework of either 45 minutes of recorded guided meditation or 30 minutes of mindfulness meditation on their own. A study was also performed[55] psoriasis patients undergoing ultraviolet B or psoralen plus ultraviolet A light treatments randomly assigned to two groups, the experimental group listening to mindfulness meditation tapes during the treatment and the control group. Patients in the mindfulness meditation tape group reached halfway clearing and full clearing significantly more rapidly than the controls did for both ultraviolet B and psoralen plus ultraviolet A treatments.

Using biofeedback in dermatology

Biofeedback can bring to conscious awareness physiologic processes that are normally out of awareness, such as breathing rate, blood pressure, muscle tension, skin temperature, and skin sweating, and can help the person to reregulate these through awareness and to develop new patterns of physiologic functioning and habitual activities. Relaxation assisted by biofeedback can have a positive effect on inflammatory and emotionally triggered skin conditions such as acne,[56] atopic dermatitis,[57] dyshidrotic dermatitis,[58] hyperhidrosis,[59] lichen planus, neurodermatitis, psoriasis,[60] and urticaria, in part through influencing immunoreactivity.[61] Patients who have low hypnotic ability may be especially suitable for this type of relaxation training.

Biofeedback devices for relaxation include galvanic skin resistance for sweating and the simple temperature sensor card. The Stress Control Card,[62] with a heat-sensitive color changing biofeedback thermometer placed on a credit card–sized card, has color indications from colder black through red and green to warmer blue. The heat-sensitive area measures ranges of finger temperatures, giving biofeedback of vasoconstriction versus vasodilation associated with autonomic activity. Finger warming can be useful for Raynaud disease.[63] Combining hypnosis with biofeedback for finger warming can have a synergistic effect.[64] Other biofeedback devices promoting relaxation include heart rate variability monitoring devices, such as the em-Wave Personal Stress Reliever and the StressEraser. Another device that coaches slowing respirations paced by music is the RespeRate. Slowing the breaths to six per minute in itself suffices to induce relaxation and shift from sympathetic to parasympathetic dominance for promoting healing.

Conclusions

Hypnosis, meditation, and biofeedback can be valuable adjuncts to dermatologic treatment. They have been underutilized and under-reimbursed in the American medical system. When conventional treatments have failed to produce sufficient results for a patient, hypnosis, meditation, or biofeedback may well be appropriate options to consider. Their relatively low costs and excellent safety profiles along with their effectiveness can make them good choices that may substantially help selected patients. Appropriate training of dermatologists in awareness of their capabilities and appropriate reimbursement could actually help to reduce medical care costs, as has been already reported in using hypnosis to relax patients for interventional radiology. Further research funding is needed to provide convincing data to insurers on the effectiveness and cost efficiencies available through the uses of hypnosis, meditation, and biofeedback.

References

1. Rossi EL. Hypnosis and ultradian cycles: A new state(s) theory of hypnosis? *Am J Clin Hypn.* 1982;25:21-32.
2. Rainville P, Hofbauer RK, Bushnell MC, et al. Hypnosis modulated activity in brain structures involved in the regulation of consciousness. *J Cogn Neurosci.* 2002;14:887-901.
3. Freeman R, Barabasz A, Barabasz M, et al. Hypnosis and distraction differ in their effects on cold pressor pain. *Am J Clin Hypn.* 2000;43:137-148.
4. Marmer MJ. *Hypnosis in Anesthesiology.* Springfield, IL: Charles C Thomas. 1959.
5. Barabasz A, Watkins JG. *Hypnotherapeutic Techniques.* 2nd ed. New York, NY: Brunner-Routledge. 2004.
6. Sulzberger MB, Wolf J. The treatment of warts by suggestion. *Med Rec.* 1934;140:552-556.
7. Ewin DM, Eimer BN. *Ideomotor Signals for Rapid Hypnoanalysis: A How-to Manual. Springfield, IL: Charles C Thomas. 2006.*
8. Shenefelt PD. Ideomotor signaling: From divining spiritual messages to discerning subconscious answers during hypnosis and hypnoanalysis, a historical perspective. *Am J Clin Hypn.* 2011;53:157-167.
9. Shenefelt PD, Shenefelt DA. Spiritual and religious aspects of skin and skin disorders. *Psychol Res Behav Manag.* 2014;7:201-212.
10. Liossi C, White P, Hatira P. A randomized clinical trial of a brief hypnosis intervention to control venipuncture-related pain of paediatric cancer patients. *Pain.* 2009;142:255-263.
11. Lang EV, Benotsch EG, Fick LJ, et al. Adjunctive non-pharmacological analgesia for invasive medical procedures: A randomised trial. *Lancet.* 2000;355:1486-1490.

12. Fick LJ, Lang EV, Logan HL, et al. Imagery content during nonpharmacologic analgesia in the procedure suite: Where your patients would rather be. *Acad Radiol.* 1999;6:457-463.
13. Shenefelt PD. Hypnosis-facilitated relaxation using self-guided imagery during dermatologic procedures. *Am J Clin Hypn.* 2003;45:225-232.
14. Shenefelt PD. Anxiety reduction using hypnotic induction and self-guided imagery for relaxation during dermatologic procedures. *Int J Clin Exp Hypn.* 2013;61:305-318.
15. Hollander MB. Excoriated acne controlled by post-hypnotic suggestion. *Am J Clin Hypn.* 1959;1:122-123.
16. Shenefelt PD. Using hypnosis to facilitate resolution of psychogenic excoriations in acne excoriée. *Am J Clin Hypn.* 2004;46:239-245.
17. Gupta MA, Gupta AK, Watteel GN. Stress and alopecia areata: A psychodermatologic study. *Acta Derm Venereol.* 1997;77:296-298.
18. Willemsen R, Vanderlinden J, Deconinck A, et al. Hypnotherapeutic management of alopecia areata. *J Am Acad Dermatol.* 2006;55:233-237.
19. Stewart AC, Thomas SE. Hypnotherapy as a treatment for atopic dermatitis in adults and children. *Br J Dermatol.* 1995;132:778-783.
20. Twerski AJ, Naar R. Hypnotherapy in a case of refractory dermatitis. *Am J Clin Hypn.* 1974;16:202-205.
21. Mason AA. A case of congenital ichthyosiform erythrodermia of Brocq treated by hypnosis. *Br Med J.* 1952;2:422-423.
22. Kidd CB. Congenital ichthyosiform erythroderma treated by hypnosis. *Br J Dermatol.* 1966;78:101-105.
23. Tobia L. L'ipnosi in dermatologia. *Minerva Med.* 1982;73:531-537.
24. Griesemer RD. Emotionally triggered disease in a dermatological practice. *Psychiatr Ann.* 1978;8:49-56.
25. Shenefelt PD. Hypnosis in dermatology. *Arch Dermatol.* 2000;136:393-399.
26. Shenefelt PD. Psychocutaneous hypnoanalysis: Detection and deactivation of emotional and mental root factors in psychosomatic skin disorders. *Am J Clin Hypn.* 2007;50:131-136.
27. Chakravarty K, Pharoah PD, Scott DG, et al. Erythromelalgia—the role of hypnotherapy. *Postgrad Med J.* 1992;68:44-46.
28. Jabush M. A case of chronic recurring multiple boils treated with hypnotherapy. *Psychiatr Q.* 1969;43:448-455.
29. Golan HP. The use of hypnosis in the treatment of psychogenic oral pain. *Am J Clin Hypn.* 1997;40:89-96.
30. Bertolino R. L'ipnosi in dermatologia. *Minerva Med.* 1983;74:2969-2973.
31. Hölzle E. Therapie der hyperhidrosis. *Hautarzt.* 1994;35:7-15.
32. Schneck JM. Ichthyosis treated with hypnosis. *Dis Nerv Syst.* 1954;15:211-214.
33. Scott MJ. *Hypnosis in Skin and Allergic Diseases. Springfield, IL: Charles C Thomas. 1960.*
34. Lehman RE. Brief hypnotherapy of neurodermatitis: A case with four-year followup. *Am J Clin Hypn.* 1978;21:48-51.
35. Iglesias A. Three failures of direct suggestion in psychogenic dermatitis followed by successful intervention. *Am J Clin Hypn.* 2005;47:191-198.
36. Ament P, Milgram H. Effects of suggestion on pruritus with cutaneous lesions in chronic myelogenous leukemia. *N Y State J Med.* 1967;67:833-835.
37. Kantor SD. Stress and psoriasis. *Cutis.* 1990;46:321-322.
38. Winchell SA, Watts RA. Relaxation therapies in the treatment of psoriasis and possible pathophysiologic mechanisms. *J Am Acad Dermatol.* 1988;18:101-104.
39. Zachariae R, Oster H, Bjerring P, et al. Effects of psychologic intervention on psoriasis: A preliminary report. *J Am Acad Dermatol.* 1996;34:1008-1015.
40. Kline MV. Psoriasis and hypnotherapy: A case report. *Int J Clin Exp Hypn.* 1954;2:318-322.
41. Frankel FH, Misch RC. Hypnosis in a case of long-standing psoriasis in a person with character problems. *Int J Clin Exp Hypn.* 1973;21:121-130.
42. Waxman D. Behaviour therapy of psoriasis—a hypnoanalytic and counter-conditioning technique. *Postgrad Med J.* 1973;49:591-595.
43. Tausk F, Whitmore SE. A pilot study of hypnosis in the treatment of patients with psoriasis. *Psychother Psychosom.* 1999;68:221-225.
44. Barabasz M. Trichotillomania: A new treatment. *Int J Clin Exp Hypn.* 1987;35:146-154.
45. Zalsman G, Haggai Hermesh H, Sever J. Hypnotherapy in adolescents with trichotillomania: Three cases. *Am J Clin Hypn.* 2001;44:63-68.
46. Perloff MM, Spiegelman J. Hypnosis in the treatment of a child's allergy to dogs. *Am J Clin Hypn.* 1973;15:269-272.
47. Shertzer CL, Lookingbill DP. Effects of relaxation therapy and hypnotizability in chronic urticaria. *Arch Dermatol.* 1987;123:913-916.
48. Noll RB. Hypnotherapy for warts in children and adolescents. *J Dev Behav Pediatr.* 1994;15:170-173.
49. Sinclair-Gieben AH, Chalmers D. Evaluation of treatment of warts by hypnosis. *Lancet.* 1959;2:480-482.
50. Spanos NP, Williams V, Gwynn MI. Effects of hypnotic, placebo, and salicylic acid treatments on wart regression. *Psychosom Med.* 1990;52:109-114.
51. Ewin DM. Hypnotherapy for warts (verruca vulgaris): 41 consecutive cases with 33 cures. *Am J Clin Hypn.* 1992;35:1-10.
52. Hautmann G, Panconesi E. Vitiligo: A psychologically influenced and influencing disease. *Clin Dermatol.* 1997;15:879-890.
53. Otani A. Eastern meditative techniques and hypnosis: A new synthesis. *Am J Clin Hypn.* 2003;46:97-108.
54. Benson H. *The Relaxation Response.* New York, NY: Morrow. 1975.
55. Kabat-Zinn J. Influence of a mindfulness meditation-based stress reduction intervention on rates of skin clearing in patients with moderate to severe psoriasis undergoing phototherapy (UVB) and photochemotherapy (PUVA). *Psychosom Med.* 1998;60:625-632.
56. Hughes H, Brown BW, Lawlis GF, et al. Treatment of acne vulgaris by biofeedback relaxation and cognitive imagery. *J Psychosom Res.* 1983;3:185-191.
57. Haynes SN, Wilson CC, Jaffe PG, et al. Biofeedback treatment of atopic dermatitis: Controlled case studies of eight cases. *Biofeedback Self Regul.* 1979;4:195-209.
58. Koldys KW, Meyer RP. Biofeedback training in the therapy of dyshidrosis. *Cutis.* 1979;24:219-221.
59. Duller P, Gentry WD. Use of biofeedback in treating chronic hyperhidrosis: A preliminary report. *Br J Dermatol.* 1980;103:143-146.
60. Benoit J, Harrell EH. Biofeedback and control of skin cell proliferation in psoriasis. *Psychol Rep.* 1980;46:831-839.
61. Tausk FA. Alternative medicine: Is it all in your mind? *Arch Dermatol.* 1998;134:1422-1425.
62. Barrios AA. *Towards Greater Freedom and Happiness. Los Angeles, CA: SPC Press. 1985.*
63. Sarti MG. Biofeedback in dermatology. *Clin Dermatol.* 1998;16:711-714.
64. Dikel W, Olness K. Self-hypnosis, biofeedback, and voluntary peripheral temperature control in children. *Pediatrics.* 1980;66:335-340.